

**2.51 describe the role of phloem in transporting sucrose and amino acids between the leaves and other parts of the plant**

- 2.52 describe the role of xylem in transporting water and mineral salts from the roots to other parts of the plant
- 2.53 explain how water is absorbed by root hair cells
- 2.54 understand that transpiration is the evaporation of water from the surface of a plant
- 2.55 explain how the rate of transpiration is affected by changes in humidity, wind speed, temperature and light intensity
- 2.56 describe experiments to investigate the role of environmental factors in determining the rate of transpiration from a leafy shoot

**Humans**

- 2.57 describe the composition of the blood: red blood cells, white blood cells, platelets and plasma
- 2.58 understand the role of plasma in the transport of carbon dioxide, digested food, urea, hormones and heat energy
- 2.59 explain how adaptations of red blood cells, including shape, structure and the presence of haemoglobin, make them suitable for the transport of oxygen
- 2.60 describe how the immune system responds to disease using white blood cells, illustrated by phagocytes ingesting pathogens and lymphocytes releasing antibodies specific to the pathogen